DHVANI – Complete Test Guide (Easy Version)

Oct 2, 2026 · @harshit

What DHVANI does

DHVANI is a sonar transmitter that changes its "ping" by itself when the water changes. It reads the water, builds the right wave, and sends it out — while the main chip sleeps to save battery.

Function	What it means in simple words	Where you see it
Wave engine	The STM32 calculates the wave and stores it in memory	Serial line shows mode and frequency
Timer + DMA	A hardware timer pushes one sample out every 1 µs; the chip does not do it by hand	Scope: CH2 stays LOW during the ping
Built-in DAC	Turns numbers into a real voltage on PA4	Scope: yellow wave
3 wave types	LFM chirp, geometric sweep, phase-coded (Barker-13)	Blue button changes them
2 bands	Audible (2–17 kHz, you can hear it) and ultrasonic (100–250 kHz, real sonar band)	Blue LED on = ultrasonic
Turbidity dial	The knob acts as the water-clarity sensor (left = clear reef, right = muddy estuary)	Serial: clarity number
Salinity (TDS)	Real probe in the water glass	Serial: tds ppm
Temperature	Real DS18B20 probe	Serial: T and sound speed c
Power meter	INA219 measures the chip's real current	Serial: I mA and E mJ
Battery saver	When the battery budget drops, pings get slower, shorter and quieter	Serial: NORMAL / ECONOMY / SURVIVE
Hann window	The ping fades in and out smoothly, no sudden jumps	Scope: swell-and-fade shape

Function	What it means in simple words	Where you see it
Op-amp filter	TL074 active low-pass + buffer smooths the DAC steps	Scope: smooth wave after the filter

What adapts, and to what:

- **Knob (turbidity) → frequency band and pulse length.** Clear = high frequency, short pulse (sharp picture). Muddy = low frequency, long pulse (goes further).
- **TDS (salt) → amplitude.** Saltier water = louder ping.
- **Battery budget → all three.** Low battery = slower, shorter, quieter pings.
- **Temperature + salt → sound speed → listening time** between pings.

Table setup (do this once)

One glass of tap water, the knob in your hand, two scope probes. The turbidity sensor is not used — the knob replaces it.

Item	Where it goes
TDS probe	In a glass of plain tap water
Temperature probe	In the same glass
Knob (potentiometer)	Left leg 3V3, middle leg A0 (CN9 pin 1), right leg GND
Salt (one spoon)	Kept ready — added only in Test 6
9 V battery	Clipped on (TL074 needs −9 V)
Scope CH1 probe tip	Free end of the 47 Ω (filtered output)
Scope CH1 ground clip	GND (CN7 pin 8)
Scope CH2 probe tip	PE0 = CN10 pin 33 (CPU-busy pin)
Scope CH2 ground clip	GND
Laptop serial console	STLink COM port, 115200

Water rule — simple:

- **Clear water vs muddy water = the KNOB.** Knob left = clear reef. Knob right = muddy estuary. You do not change the glass.
- **Fresh water vs salty water = the GLASS.** Plain tap water first. Add salt only in Test 6.

Before every test session:

1. Press black RESET on the board → battery budget 100% NORMAL.
2. Knob in the middle.
3. Mode AUD-LFM (blue LED off). Blue button order: AUD-LFM → AUD-GEO → AUD-BPSK → US-LFM → US-GEO → US-BPSK.

Scope base settings:

Setting	Value
CH1 probe switch + CH1 menu Probe	×10 and 10×
CH1 coupling / scale	DC, 500 mV/div (AC only for FFT)
CH2 probe switch + CH2 menu Probe	×10 and 10×
CH2 coupling / scale	DC, 2 V/div, trace in lower half
Trigger	Edge, Source CH2, Slope falling, Level 1.6 V (push LEVEL knob = 50%), Sweep Normal

Why you don't see 100–500 kHz

You were in the audible band, so the scope showed 2–17 kHz. The ultrasonic band appears only in the US modes, and it reaches 120–250 kHz, not 500 kHz.

Reason 1 — the mode. The first three modes (AUD-LFM, AUD-GEO, AUD-BPSK) are deliberately audible so the buzzer and your phone can hear them. Press the blue button 3 times → blue LED on → US-LFM. Now the band is 100 kHz up to 120–250 kHz.

Reason 2 — the scope setting. At 1–2 ms/div the scope samples only 25–50 kSa/s, so it cannot show anything above 12.5–25 kHz. For ultrasonic you must zoom in: 5 µs/div for the wave, 50–100 µs/div for the FFT.

Reason 3 — the hardware limit. The built-in DAC runs at 1 million samples per second. A clean sine needs at least about 4 samples per cycle, so the top clean frequency is about 250 kHz. 500 kHz would be only 2 samples per cycle — a rough triangle, not a sine.

Band	Shown on	What you get
100–250 kHz	US modes, scope at 5 µs/div	Clean chirp, adapts with the knob
250–500 kHz	Not reachable cleanly	Needs a faster external DAC (≥ 2 MSps)

What to tell the judges: "The brief gives 100–500 kHz as example bands. Our prototype covers 100–250 kHz with the MCU's internal 12-bit DAC at 1 MSps, adapting live to water clarity. A faster external DAC on the December PCB extends this to 500 kHz."

All tests, step by step

Nine tests, about 45 minutes. Take a photo after each one — these become your results slides.

Test 0 — Serial check (no scope)

Mode AUD-LFM · knob middle · plain tap water.

1. Open the serial console. One line appears every ping.
2. Turn knob fully left → `clarity` ~4000, `f=2000-17000Hz`, `dur=4ms`.
3. Turn knob fully right → `clarity` ~0, `f=2000-5000Hz`, `dur=16ms`.
4. Hold the temperature probe in your hand → `T` rises, `c` rises.
5. Copy one line at each knob end into a text file.

Brief: sensors read via ADC; adaptation happens instantly.

Test 1 — CPU sleeps during the ping

Mode AUD-LFM · knob middle · plain water · CH1 + CH2 on · 5 ms/div.

1. Look at the green CH2 line.
2. Expect: green HIGH (CPU building the wave), then LOW for the whole yellow burst, then a tiny spike after it.

Brief: timers and DMA stream the wave "without stalling the CPU". Already passed — your first photo.

Test 2 — Smooth ping shape (Hann window)

Mode AUD-LFM · knob middle · CH2 off · 2 ms/div.

1. Expect: a burst that swells and fades, flat 1.65 V line before and after.

Brief: digital window removes sudden voltage jumps at start and end.

Test 3 — Three modulation types

Knob middle · 500 µs/div · press Single, scroll with Horizontal Position.

1. AUD-LFM → cycles get narrower at a steady rate.
2. Blue button once → AUD-GEO → narrows slowly, then fast at the end.

3. Blue button again → AUD-BPSK → same width all through, with small kinks where the phase flips.
4. Press Run/Stop after each photo.

Brief: "LFM chirps, geometric sweeps, and phase-coded pulses" on the fly. Already passed — your photos.

Test 4 — FFT of each type

Knob middle · CH1 coupling AC · 1 ms/div (Sa = 50 kSa) · MATH → FFT, Source CH1, Hanning, dB, Display Split.

1. Read across: each square = 2.5 kHz, left edge = 0 Hz.
2. AUD-LFM → raised plateau from ~2 kHz to ~11 kHz.
3. AUD-GEO → band taller on the low side, sloping down.
4. AUD-BPSK → one hump in the middle (~6–7 kHz).
5. A small extra hump at ~20+ kHz is a harmonic — ignore it.

Brief: clean, mathematically sound spectrum via FFT.

Test 5 — Knob changes frequency and pulse length

Mode AUD-LFM · plain water.

1. FFT on (same as Test 4). Knob fully LEFT (clear reef) → plateau's right edge moves out to ~17 kHz (about 7 squares). Photo.
2. Knob fully RIGHT (muddy estuary) → right edge moves in to ~5 kHz (2 squares). Photo.
3. FFT off, CH1 back to DC, 5 ms/div, Cursors → Time.
4. Knob LEFT → burst ~4 ms long. Photo.
5. Knob RIGHT → burst ~16 ms long. Photo.

Brief: parameter 1 (bandwidth/centre frequency) and parameter 2 (pulse duration) change with the environment. Clear = high frequency + short pulse for sharp images. Muddy = low frequency + long pulse to penetrate.

Test 6 — Salt changes amplitude

Mode AUD-LFM · knob middle · 2 ms/div · Measure → Vpp.

1. Plain tap water → note Vpp (about 1.9–2.0 V). Photo.
2. Stir one spoon of salt into the same glass, wait 5 s.
3. Salty water → Vpp rises (up to about 2.9 V). Photo.
4. Check: Vpp ≈ 2 × amp × 3.3 ÷ 4095, using `amp` from serial.

Brief: parameter 3 (amplitude / signal power) changes with the environment.

Test 7 — Before vs after the op-amp filter

Mode US-LFM (blue button 3 times, blue LED on) · knob middle · 5 µs/div · Single.

1. Move CH1 tip to PA4 (CN7 pin 9) → staircase steps. Photo.
2. Move CH1 tip back to the 47 Ω point → smooth sine. Photo.
3. MATH → FFT, 50 µs/div (span ~0–1.5 MHz). At PA4 → spike near 1 MHz. Photo.
4. At the 47 Ω point → that spike much smaller. Photo.

Brief: active low-pass filter + operational amplifier smooth the DAC output.

Test 8 — Ultrasonic band adapts

Mode US-LFM · FFT on · 100 µs/div (span ~0–300 kHz).

1. Knob LEFT (clear) → band reaches ~250 kHz. Photo.
2. Knob RIGHT (muddy) → band reaches only ~120 kHz. Photo.

Brief: high frequency for clear water, low frequency for murky water — in the real sonar band.

Test 9 — Battery saver

Any mode · watch serial · leave running about 10 minutes.

1. `batt` drops from 100%.
2. Below 50% → ECONOMY: amp × 0.8, duration × 0.75, slower pings.
3. Below 20% → SURVIVE: amp × 0.6, duration × 0.5, much slower pings.
4. RESET → back to 100% NORMAL.

Brief: does not drain the AUV's battery; power measured live by INA219 (~12 mA).

Brief requirement → which test proves it

Every line of the PS 26058 brief is covered by at least one test above; only the enclosure is still to do.

Brief requirement	Proved by	What the judge sees
Physical hardware on an embedded platform	All	STM32U575 Nucleo + breadboard analog front end

Brief requirement	Proved by	What the judge sees
Timers + DMA to DAC without stalling the CPU	Test 1	CH2 LOW during the whole ping
LFM, geometric sweep, phase-coded on the fly	Test 3, Test 4	Three different waves and spectra, one button
Sensors / dials read via ADC	Test 0	Knob, TDS, temperature on serial
1. Bandwidth / centre frequency adapts	Test 5, Test 8	FFT band moves with the knob
2. Pulse duration adapts	Test 5	Burst 4 ms (clear) vs 16 ms (muddy)
3. Amplitude / power adapts	Test 6	Vpp rises with salt
Digital windowing (Hann)	Test 2	Smooth swell and fade, no jumps
Active low-pass filter + op-amp	Test 7	Staircase at PA4, smooth after TL074
Clean FFT validation	Test 4, Test 7	Clean bands, DAC image suppressed
Low power, battery-aware	Test 1, Test 9	~12 mA, mJ per ping, ECONOMY / SURVIVE
100 kHz (murky) vs high frequency (clear)	Test 8	120 kHz muddy vs 250 kHz clear
3D-printed payload pod	—	Still to build

Honest notes for the slides:

- Turbidity is simulated with a dial, as the brief allows; TDS, temperature and power are real sensors.
- Ultrasonic band reached: 100–250 kHz; 500 kHz is the December PCB step.

Scope quick reference and fixes

The DSO1052B buttons you need, and what to do when the screen looks wrong.

Button / knob	What it does
Single	Freeze one ping
Run/Stop	Un-freeze
Horizontal Position knob	Scroll along a frozen ping

Button / knob	What it does
Time/div knob	Zoom in time; in FFT it also sets the frequency range
MATH (between CH1 and CH2 buttons)	FFT on/off and settings
Cursors	Measure burst length (ΔT)
Measure	Vpp and frequency
Trigger LEVEL knob	Push it = level set to the middle automatically
Auto	Emergency reset if the screen is blank
Save/Recall → Bitmap → USB	Save a screenshot

Time/div	Sample rate	FFT can show up to	Use for
2 ms	25 kSa	12.5 kHz	Ping shape only, not FFT
1 ms	50 kSa	25 kHz	Audible FFT (Tests 4, 5)
100 µs	500 kSa	250 kHz	Ultrasonic FFT (Test 8)
50 µs	1 MSa	500 kHz+	Filter FFT (Test 7)
5 µs	—	—	Ultrasonic wave detail (Test 7)

Problem	Fix
Picture keeps scrolling	Trigger Sweep = Normal, Source CH2, Slope falling
Wave 10× too big or small	Probe switch and channel menu must both say ×10
Flat line	Check ground clip and tip; press Auto
Extra hump on the right of the FFT	Sample rate too low — reduce Time/div
Big spike at FFT left edge	CH1 coupling AC
FFT partly off-screen	Press MATH, then move it up with the vertical position knob; or Display Split
Serial says SURVIVE, ping quiet	Press RESET on the board
No change when turning the knob	Middle leg must be on A0; USE_DIAL must be 1